Year 10 Higher

REVISION PACK

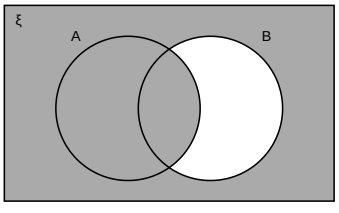
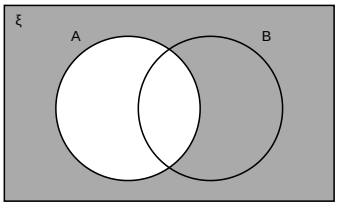
Sets 1

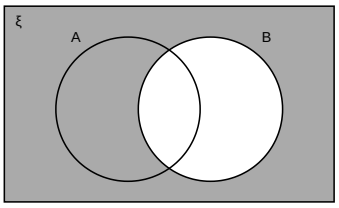
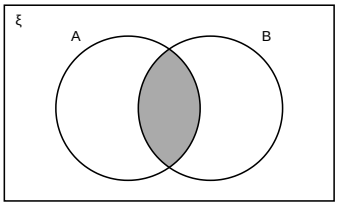
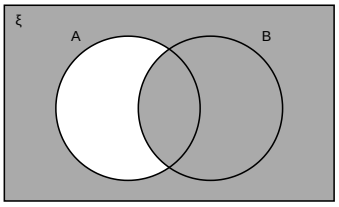
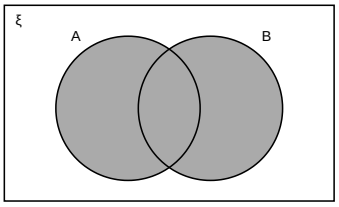
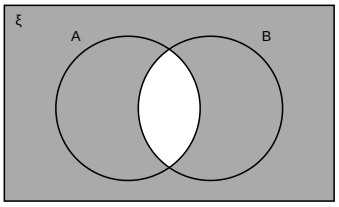
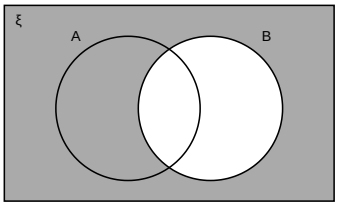
ANSWERS

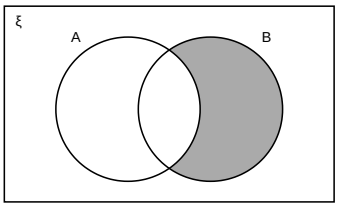
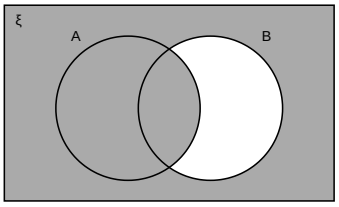
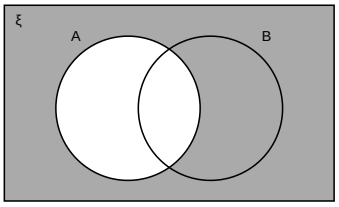
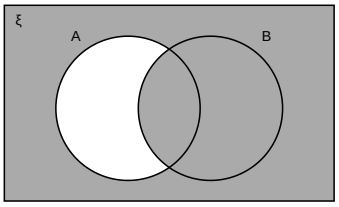
Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Q	Working	Answer
listing the members of a set		
1.1		{2, 3, 5, 7, 11, 13, 17, 19, 23, 29}
1.2		{1, 2, 3, 4, 6, 9, 12, 18, 36}
1.3		{14, 21, 28, 35, 42, 49, 56}
1.4		{1, 8, 27, 64}
1.5		{1, 2, 3, 5, 6, 10, 15, 30}
1.6		{21, 23, 25, 27, 29, 31}
intersection, union and complement		
1.7	(a) $A = \{3, 6, 9, 12\}$ $B = \{1, 2, 3, 4, 6, 12\}$ (b) $A \cup B = \{1, 2, 3, 4, 6, 9, 12\}$ (c)	(a) {3, 6, 12} (b) 7 (c) Yes: 7 is in neither A nor B
1.8	(a) $A = \{1, 3, 5, 7, 9, 11\}$ $B = \{1, 2, 4, 8\}$ (b) $A \cup B = \{1, 2, 3, 4, 5, 7, 8, 9, 11\}$ (c)	(a) {1} (b) 9 (c) Yes: 6 is in neither A nor B
1.9	(a) $A = \{5, 10\}$ $B = \{2, 3, 5, 7, 11\}$ (b) $A \cup B = \{2, 3, 5, 7, 10, 11\}$ (c)	(a) {5} (b) 6 (c) Yes: 9 is in neither A nor B
1.10	(a) $A = \{1, 4, 9\}$ $B = \{1, 2, 3, 6\}$ (b) $A \cup B = \{1, 2, 3, 4, 6, 9\}$ (c)	(a) {1} (b) 6 (c) No: 4 is in A

1.11	**(a)** $A = \{4, 8, 12\}$ $B = \{1, 2, 4, 8\}$ **(b)** $A \cup B = \{1, 2, 4, 8, 12\}$ **(c)**	**(a) {4, 8}** **(b) 5** **(c) Yes: 3 is in neither A nor B**
1.12	**(a)** $A = \{1, 4, 9\}$ $B = \{3, 6, 9, 12\}$ **(b)** $A \cup B = \{1, 3, 4, 6, 9, 12\}$ **(c)**	**(a) {9}** **(b) 6** **(c) No: 12 is in B**
three-set Venn diagrams		
1.13	$(a)\ x = 45 - 37 = 8$ $(b)\ 3 + 2 = 5$ $(c)\ 45 - (5 + 4 + 6 + 2) = 28$	**(a)** $x = 8$ **(b) 5** **(c) 28**
1.14	$(a)\ x = 40 - 34 = 6$ $(b)\ 5 + 1 = 6$ $(c)\ 40 - (10 + 2 + 3 + 1) = 24$	**(a)** $x = 6$ **(b) 6** **(c) 24**
1.15	$(a)\ x = 60 - 47 = 13$ $(b)\ 6 + 5 = 11$ $(c)\ 60 - (8 + 3 + 4 + 5) = 40$	**(a)** $x = 13$ **(b) 11** **(c) 40**
1.16	$(a)\ x = 35 - 32 = 3$ $(b)\ 2 + 1 = 3$ $(c)\ 35 - (7 + 5 + 3 + 1) = 19$	**(a)** $x = 3$ **(b) 3** **(c) 19**
shading regions		
1.17	(a) $A \cup B'$  (b) A'  	

1.18	(a) B'  (b) $A \cap B$  	
1.19	(a) $A' \cup B$  (b) $A \cup B$  	
1.20	(a) $(A \cap B)'$  (b) B'  	

1.21	(a) $A' \cap B$  (b) $A \cup B'$  	
1.22	(a) A'  (b) $A' \cup B$  	
problems with two sets		
1.23	**(a)** $22 + 16 + 6 = 44$ $44 - 40 = 4$ **(b)** $22 - 4 = 18$	**(a) 4** **(b) 18**
1.24	**(a)** $28 + 25 + 5 = 58$ $58 - 50 = 8$ **(b)** $28 - 8 = 20$	**(a) 8** **(b) 20**
1.25	**(a)** $20 + 14 + 8 = 42$ $42 - 35 = 7$ **(b)** $20 - 7 = 13$	**(a) 7** **(b) 13**

1.26	**(a)** $25 + 20 + 10 = 55$ $55 - 45 = 10$ **(b)** $25 - 10 = 15$	**(a) 10** **(b) 15**
1.27	**(a)** $16 + 12 + 6 = 34$ $34 - 30 = 4$ **(b)** $16 - 4 = 12$	**(a) 4** **(b) 12**
1.28	**(a)** $35 + 30 + 5 = 70$ $70 - 60 = 10$ **(b)** $35 - 10 = 25$	**(a) 10** **(b) 25**